CASE STUDY

Process Planning for a Manufactured Component

A manufacturing process plan for a machined flanged hub

Component selected: Machined flanged hub

Manufacturing route: Cutting, turning, drilling, milling, inspection and packing

Report type: Manufacturing engineering case study

Prepared as an academic case report

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1 Introduction

Process planning converts a component drawing into a repeatable sequence of manufacturing operations. The planner decides the starting stock, machines, tools, fixtures, inspection points and operating conditions that will produce the required geometry at an acceptable cost. This case study develops a practical route for a machined flanged hub used to transmit torque between a shaft and a bolted coupling.

The selected component combines rotational features, a precision bore, a keyway and a bolt circle. It is therefore suitable for showing how turning, drilling and milling operations are coordinated. The resulting plan is intended for a small to medium batch-production workshop equipped with conventional CNC turning and vertical machining capability.

2 Background and Problem Statement

A flanged hub must fit a mating shaft without excessive clearance, locate concentrically with a coupling flange and transfer torque through a parallel key. Poor process planning can cause runout between the bore and outer diameter, burrs in the keyway, inconsistent bolt-circle location or unnecessary handling time. These faults increase rework and can lead to vibration, assembly difficulty and early service failure.

The problem is to prepare a manufacturing process plan that meets the drawing requirements while reducing set-ups, maintaining a clear datum scheme and building inspection into the route. The plan must be understandable to production operators and sufficiently robust for a batch size of 250 parts.

3 Objectives and Success Criteria

Objective	Success criterion
Establish an economical route	No more than three major machine set-ups after cutting.
Control critical geometry	Bore 30 H7 and bore-to-OD runout not greater than 0.03 mm.
Protect functional surfaces	No visible burrs at the bore, keyway or threaded holes.
Support batch production	Documented tools, datums and in-process inspection for every batch.
Improve traceability	First-off and final inspection records linked to the batch number.

4 Component Description and Production Assumptions

The component is a flanged hub made from normalized EN8/C45 medium-carbon steel. The hub has a stepped outside diameter, a precision through bore and a radial flange with six mounting holes. Dimensions below are representative process-planning inputs; the engineering drawing remains the final authority for manufacture.

Feature	Nominal requirement	Process significance
Starting stock	90 mm diameter bar x 95 mm long	Allows facing and turning allowance.
Flange OD	84.00 mm	Concentric with precision bore.
Hub OD	56.00 mm	Turned in same chucking as bore.
Overall length	78.00 mm	Controlled by faced datum surface.
Bore	30 H7 through	Critical shaft fit; finish boring/reaming.
Keyway	8 mm wide x 3.3 mm deep	Milled after bore is established.
Bolt circle	6 x M8 on 68 mm PCD	Requires indexed drilling/tapping fixture.

5 Process Planning Methodology

5.1 Datum strategy

Datum A is the finished flange face. Datum B is the finished bore axis. These datums are created early because they control functional assembly. Turning the flange face, hub diameters and bore in one CNC-lathe set-up minimises concentricity error. The finished bore is then used on an expanding mandrel for keyway and bolt-circle machining.

5.2 Route selection

The route follows the principle of producing general geometry before precision geometry, and completing heavy metal-removal operations before finishing and inspection. Cutting is separated from precision machining so saw distortion and rough end condition do not influence the lathe datum. Secondary milling occurs only after the bore can locate the part accurately.

5.3 Process flow

1	2	3	4	5	6	7	8	9
Incoming inspection	Saw cutting	CNC turning and boring	Mandrel locating	Keyway milling	PCD drilling and tapping	Deburr and wash	Final inspection	Oil and pack

6 Detailed Process Plan

Op.	Operation	Machine and tooling	Key instruction
10	Receive and inspect bar stock	Material certificate; 90 mm bar; visual check	Verify grade, diameter and straightness before issue.
20	Saw cut blanks	Automatic band saw; length stop	Cut to 95 mm plus 2 mm allowance; identify batch.
30	Rough turning	CNC lathe; three-jaw chuck; CNMG insert	Face first end, rough turn ODs and pilot drill bore.
40	Finish turning and boring	Same CNC set-up; boring bar; reamer	Finish Datum A, flange/hub ODs, length and 30 H7 bore.
50	Chamfer and handling check	CNC lathe; 45 degree tool	Break sharp edges; protect finished bore during transfer.
60	Set on expanding mandrel	VMC; expanding mandrel locating on bore	Locate on Datum B and support against Datum A.
70	Mill keyway	VMC; 8 mm end mill; gauge pin	Mill 8 mm keyway to depth; inspect width and depth.
80	Drill and tap bolt circle	VMC; probe; 6-position program	Spot drill, drill 6.8 mm, tap M8 on 68 mm PCD.
90	Deburr and wash	Deburring station; aqueous washer	Remove burrs; ensure threads, bore and keyway are clean.
100	Final inspection and pack	CMM or gauges; VCI paper	Record critical dimensions; apply rust preventive and pack.

6.1 Estimated Cycle Time and Capacity

Activity	Estimated time per part	Planning note
Saw cutting	1.2 min	Batch cutting with automatic length stop.
Turning and boring	6.5 min	Includes loading, roughing, finishing and bore gauging.
Milling, drilling and tapping	5.8 min	CNC program combines keyway and bolt-circle operations.
Deburring, wash and inspection	2.0 min	Includes sampling and final clean.
Total planned direct time	15.5 min	Set-up time is planned separately per batch.

At 15.5 minutes direct time, one 250-piece batch requires approximately 64.6 direct machine-hours before allowing for set-up, tool changes, inspection and planned downtime. This estimate enables the planner to schedule capacity and compare alternative routing choices.

7 Quality Control and Risk Management

Control point	Method and frequency	Acceptance requirement	Reaction if nonconforming
Raw material	Certificate check and visual inspection for each batch	EN8/C45 grade; no severe corrosion or bends	Quarantine and notify stores/quality.
First-off turning	Vernier, bore gauge, dial indicator	Dimensions per drawing; runout <= 0.03 mm	Adjust offset or jaws; repeat first-off.
Keyway	Go/no-go gauge and depth micrometer every 25 parts	8 mm width and specified depth	Stop VMC operation; inspect last accepted interval.
PCD holes and threads	CMM/sample fixture and thread plug gauge every 25 parts	Correct 68 mm PCD; M8 thread passes gauge	Correct program/tool; rework only if approved.
Final inspection	100% visual; critical-dimension sampling	Clean, burr-free, traceable and protected	Segregate; raise nonconformance report.

7.1 Principal risks and controls

The dominant technical risk is loss of concentricity between the bore and turned diameters. It is controlled by completing both features in the same lathe set-up and verifying runout during first-off inspection. A second risk is a misplaced keyway or bolt circle. Using the finished bore on an expanding mandrel establishes a repeatable axis and avoids locating on an unfinished outside diameter. Burrs are controlled by specified chamfering, a dedicated deburring step and visual inspection before packing.

8 Conclusion and Future Scope

This case study demonstrates how a component drawing can be translated into a controlled manufacturing route. The plan starts with suitable bar stock, establishes the functional flange face and bore as datums, completes rotational features in one lathe set-up and uses the finished bore for accurate secondary machining. Inspection is placed at material receipt, first-off approval, in-process sampling and final release, creating a practical balance of quality and production efficiency.

For higher volumes, the route can be improved through bar-fed turning, automated part transfer, in-machine probing and a dedicated hydraulic fixture. Statistical process control on the bore and runout dimensions would further reduce inspection effort once the process capability is demonstrated. The same planning logic can be adapted to other shaft-coupling components with different materials, tolerances or production volumes.

Reference

Reference structure: supplied case-report PDF, used as a guide for academic report organisation. Technical dimensions and manufacturing plan in this document are independently prepared for the selected flanged hub case.